

Preserving Molecular Information During Pathology Report De-identification: A Synthetic Benchmark

Background: De-identified pathology reports could support report standardization, summarization, hypothesis generation and testing, and research abstraction. These uses require preserving molecular information while removing identifiers. We evaluated failures at this boundary.

Design: An institution-developed pipeline and context-aware rules were compared with five frontier LLM configurations (GPT-5.6 Sol through two interfaces, Claude Sonnet 5, Gemini 3.1 Pro, and Grok 4.6) on 200 unique synthetic report pairs. Success required preserving a molecular expression in one report and removing similarly formatted identifying digits in its paired control. A separate 200-pair experiment evaluated local refinement across eight report patterns. Paired analyses accounted for repeated patterns and multiple comparisons.

Results: The institution-developed pipeline preserved complete molecular expressions in 0/200 reports and exposed identifying digits in 11/200 controls. Context-aware rules preserved 159/200 expressions, exposed digits in 30/200 controls, and achieved both objectives in 133/200 pairs. LLM configurations preserved 200/200 expressions and achieved both objectives in 197-200/200 pairs, with 0-3/200 controls exposing digits (adjusted $p=0.16$ versus context-aware rules, accounting for report patterns). Local failures included molecular over-redaction and identifiers missed across line breaks, repeated footers, parenthetical fields, and modified labels. In the separate experiment, refinement reduced exposures from 100/200 to 50/200 while preserving all 200 molecular expressions; parenthetical-field and modified-label failures persisted.

Conclusions: Local refinement corrected specific failures but left substantial identifier exposure. Frontier LLM configurations performed better descriptively on this synthetic benchmark. Evaluations should measure identifier removal and molecular preservation together; these findings do not establish comprehensive anonymity or clinical generalizability.

The gene survived. The variant did not.

A familiar pathology example: BRAF p.V600E became BRAF p.***** after redaction.

SOURCE

Original synthetic report

MOLECULAR PATHOLOGY

Name: Mivin Ravven
MRN: 14727160

Specimen: Skin, shoulder
Block: A1

RESULT

BRAF p.V600E is detected.

VARIANT LOST

Institution-developed pipeline

MOLECULAR PATHOLOGY

Name: *****
: **

Specimen: Skin, shoulder
Block: A1

RESULT

BRAF p.***** is detected.

VARIANT PRESERVED

Earlier context-aware method

MOLECULAR PATHOLOGY

Name: *****
: **

Specimen: Skin, shoulder
Block: A1

RESULT

BRAF p.V600E is detected.

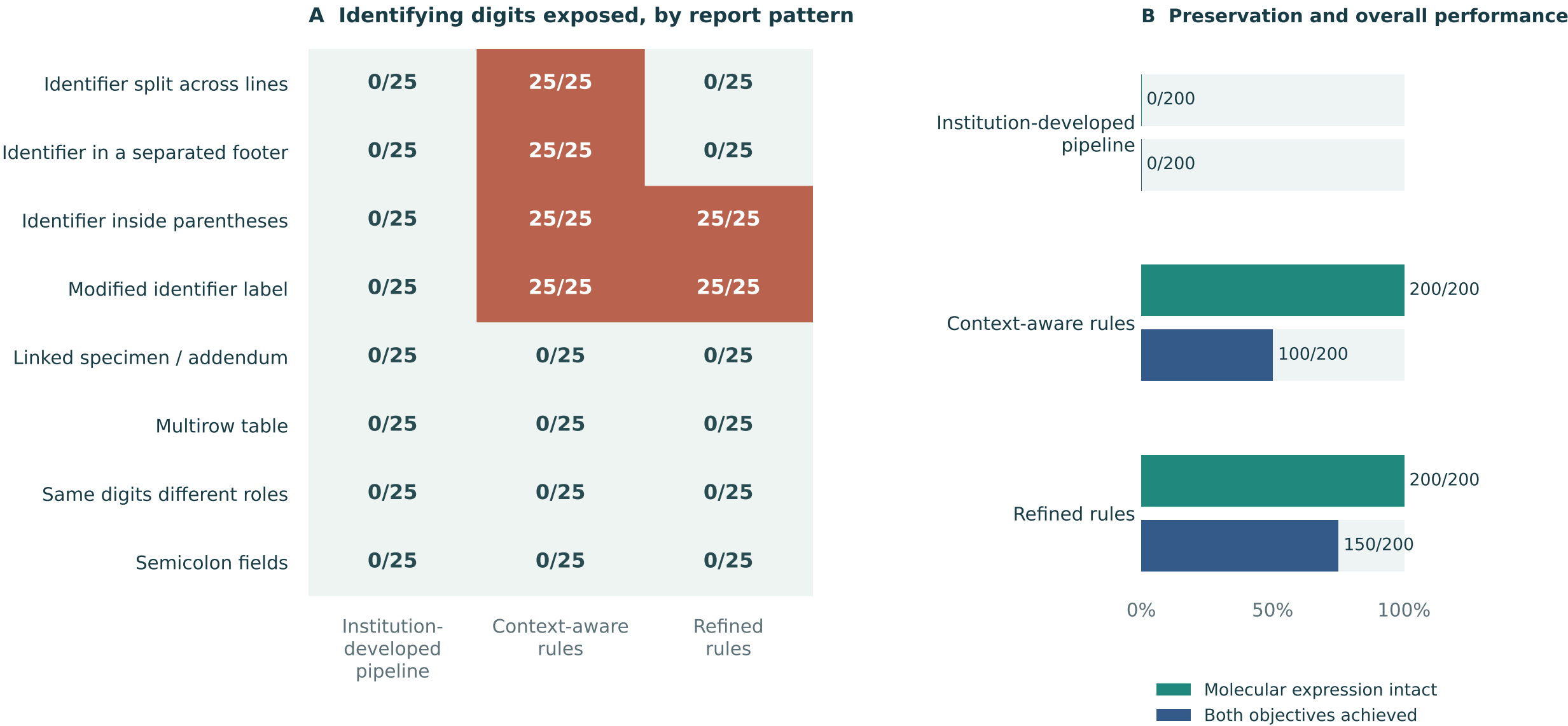
Why this matters

The output still says a BRAF alteration was detected, but its specific protein change is unreadable.
The earlier context-aware method retained the variant; that does not establish complete identifier removal.

Actual archived synthetic input and outputs · doc_000462 · selected excerpts; full report available in the viewer.
Historical illustration, outside both 200-pair experiments. It is not evidence of a measured downstream clinical error.

Local refinement: improvements and persistent failures

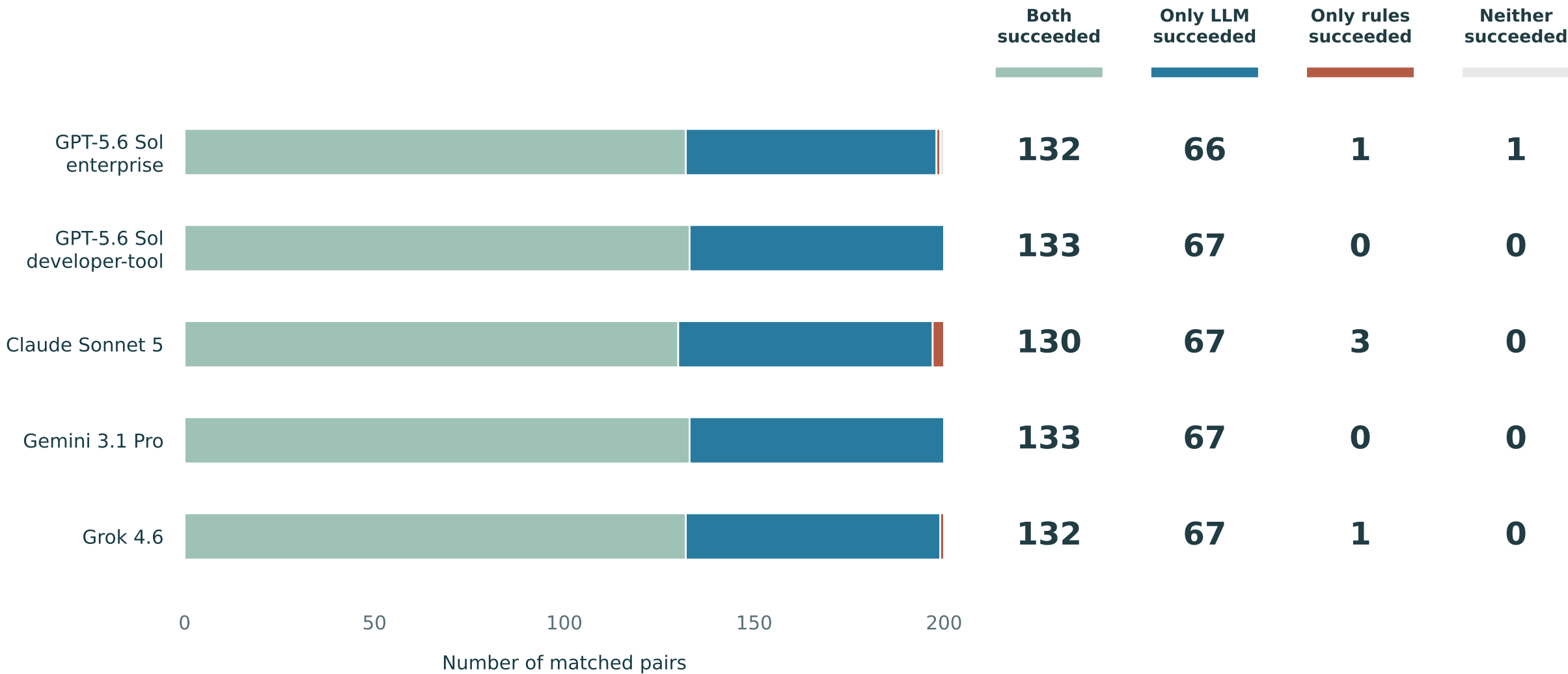
Separate experiment: 200 unique synthetic pairs · eight report patterns · 25 pairs per pattern



Zero tested-digit exposure alone does not imply success: the institution-developed pipeline preserved no complete molecular expressions. Refinement produced 50 additional successful pairs out of 200, across two report patterns (p = 0.50 accounting for patterns).

Which method succeeded on the same report pairs?

Each row contains the same 200 pairs. Success means both identifier removal and molecular preservation.



READ ONE ROW

GPT enterprise: both succeeded on 132 pairs; only GPT on 66; only local rules on 1; neither on 1.

All comparisons use context-aware local rules. Adjusted p = 0.16 after accounting for shared report patterns.
Counts are descriptive; this synthetic comparison does not establish clinical superiority. Complete statistical methods are in the supplement.